

SIMULATED DEMO COMPILED SUBSTITUTE

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Title: Scene-Memory Guided Fast-Slow Video Anomaly Detection
with Selective Vision-Language Verification

Abstract:

We study inference-cost control for fixed-camera VAD using lightweight screening, scene-normal memory, and selective VLM review. On synthetic camera-demo, Ours reaches AUROC 87.1% at 12% VLM call rate vs Lightweight 81.2% and VLM-all 100% calls. ALL METRICS ARE SIMULATED.

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Method

Pipeline: sample -> fast_score(s,u) -> should_review -> retrieve_memory
-> verify_with_vlm -> fuse -> event.

I1 trigger: $s \geq 0.65$ or $u \geq 0.2$

I2 memory: topK=5 train-normal only

I3 VLM: simulated JSON evidence; fusion weights 0.4/0.2/0.4

Figures: figures/architecture.png, figures/comparison.png
(valid PNGs present; not GPT originals).

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Results

Method	AUROC	AP	F1	VLM%	ms
Lightweight	81.2	57.8	61.4	0	18
VLM-all	85.9	64.1	66.2	100	680
FastSlow-I1	84.8	62.5	65.1	18	137
MemoryFastSlow-I1I2I3	87.1	67.2	68.0	12	99

Ablation AUROC: 81.2 -> 84.8 -> 85.6 -> 87.1

Seeds 11/23/47 Ours AUROC mean+/-std: 87.10 +/- 0.30

Conclusion: teaching demo only; real measurement required for claims.